

Data Materialized: An Introduction to a Creative Workflow

Translating Data into Form

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ABSTRACT

This hands-on course will guide participants through a creative workflow we call data materialization. Data materialization is a sub-genre of data art/data physicalization that utilizes data connected to a concept to assist in generating a 3D form to be fabricated in a meaningful material. Participants will be introduced to computational design fundamentals and foundational elements and principles of design, guided through a workflow that begins by collecting data and culminates in the design and fabrication plans of a designed data object. Tapping into the universal feeling of worry, participants in this course will design meaningful worry bead designs from associated data.

KEYWORDS

Making, Materialization

ACM Reference Format:

Courtney Starrett and Susan Reiser. 2024. Data Materialized: An Introduction to a Creative Workflow Translating Data into Form. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH Labs '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641236.3664420>

1 AUDIENCE

This course is introductory and is suitable for any attendee who is interested in making a meaningful object through a data materialization [Starrett et al. 2018] workflow and is willing to install the freely available software they will need. Students with experience in programming, modeling, and/or vector graphics will be able to customize our program or model for a more unique design. Beginners can run the code we provide without modifications, following along with each step.

2 INTRODUCTION

Data materialization uses data as a material. Creating art directly from data provides an opportunity to embed a concept or message in a form. It can communicate information about a relevant issue of our time—e.g. climate, racism, sexism—by transforming related data into a meaningful and tangible form. During the workshop we will present our workflow which uses abstract data. Abstract

data lacks a “natural notion of position in space” as opposed to data with an existing spatial layout such as weather models or medical imagery, which is often referred to as physical data [Moere 2005].

Data materializations can be realized on a spectrum of abstraction, differing from data visualizations which intend to present data in a more understandable form. Data materializations are created not with the intention to present data in a decodable format, but to conceptually connect data and the material of fabrication. Some may be more direct, abstracted only slightly, while others occupy a more non-objective location on the spectrum.

After deciding on a concept or message, we collect and curate the data which is read as an input file by our customizable Processing sketch. The program generates a 2D drawing representative of the data. A section of the vector generation of workflow is illustrated in figure 1. The vectors are imported into a 2D vector graphics program to develop cutting paths, or into 3D modeling software to serve as the basis for a 3D form. Finished 2D shapes and 3D forms can be output or fabricated using any appropriate digital fabrication technologies such as laser cutting, 3D printing, or CNC (computer numerical control) machining.

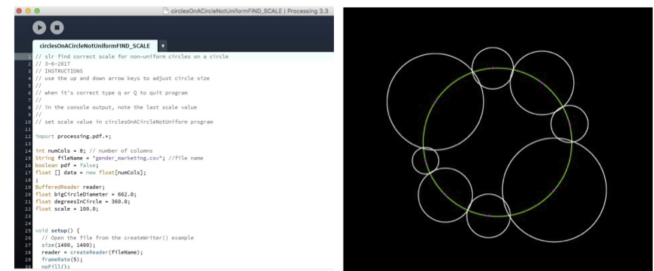


Figure 1: Processing sketch code with data generated vector output.

For SIGGRAPH 2024 we will walk participants through a data materialization workflow. We will demonstrate how to transform our worries into form and convey meaning through material. We will provide pre-sourced data sets to manifest as beads and encourage participants to source their own worry-relevant statistical data. The sample data sets will be related to the global issues of our time: climate change, political instability, and themes of diversity. Beads have a ubiquitous presence throughout human history. They transcend culture, as does worry. Creating custom worry beads seems particularly appropriate in our stressful not-quite-post-pandemic world. This lab provides participants the opportunity to create their own set of worry beads, richly symbolic objects with deep historical significance. For example, beads may be markers of civilization’s

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SIGGRAPH Labs '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0518-2/24/07

<https://doi.org/10.1145/3641236.3664420>

most advanced technologies, modes of symbolic communication, and/or currency [Dubin 2009]. Beads have even been credited for helping to advance our development of language [Pagel 2015].

In the hands-on class we will:

- Introduce participants to the data materialization workflow
- Walk participants through the steps of the data materialization workflow with pre-established codes and models
- Briefly introduce the elements and principles of design and how to evaluate a successful combination of form and meaning
- Guide participants as they finalize their designs and export STL files for fabrication
- Present options for fabrication through service bureaus and local makerspaces

Additionally, we have presented iterations of this workshop at colleges and professional conferences since 2015 as well as an outreach workshop for middle school students, Computational Bead Design, developed to , to expose the students to the foundational concepts of computer science, art, and design [Doman et al. 2014].

We have developed a tutorial and methodology for the course that will fit within a 90-minute time frame, allowing participants to customize the process either in the course or afterwards. We are confident that this version of our scripted course is scalable.

3 TIME BREAKDOWN

The 90-minute hands-on class schedule will be as follows. Time units are in minutes.

- 00:25 Introduction to Data Materialization and design
- 00:45 The hands-on process
- 00:50 Concept (choose from one of three provided data sets)
- 00:05 Formatting the data (Excel)
- 00:15 Creating the design (Fusion 360)
- 00:10 Updating the parameters to contain the data
- 00:15 Editing the 3D design (Fusion 360)
- 00:10 Small Group Discussion and Idea development
- 00:10 Fabrication and Next Steps

4 OUTCOMES/CONCLUSION

After attending the workshop, participants will be able to:

- Create original art object utilizing data as a material
- Understand the principles and elements of 3D design as applied to data materialization
- Apply a conceptual design as data materialization
- Generate 3D models from data sources
- Evaluate fabrication options and material significance / meaning

This hands-on workshop will walk participants through a data materialization workflow using data as a raw material in the creation of meaningful physical objects—worry beads. At the completion of this workshop, participants will have created a personalized design for a set of worry beads with both an aesthetic allure and an intellectual intrigue. They will leave with their design saved as an STL file that they can fabricate or have fabricated. If applicable, files can be fabricated on site in the Labs if there is a small fabrication workspace. We can adjust materials and processes to fit many different situations including but not limited to: 3D prints, paper, clay, fiber, found materials, etc.



Figure 2: EPA data on Recycled Materials by type in silver (2024).

ACKNOWLEDGMENTS

We would like to acknowledge support from Texas A&M University's Academy of Visual and Performing Arts Research Grant.

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